

Exercise of Supervised Learning: Advanced Risk Minimization Part 1

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Exercise 1: Risk Minimizers for Generalized L2-Loss

Consider the regression learning setting, i.e., $\mathcal{Y} = \mathbb{R}$, and assume that your loss function of interest is $L(y, f(\mathbf{x})) = (m(y) - m(f(\mathbf{x})))^2$, where: $m : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous strictly monotone function.

Disclaimer: In the following we always assume that $\text{Var}(m(Y))$ exists.

(a) Consider the hypothesis space of a constant models

$\mathcal{H} = \{f : \mathcal{X} \rightarrow \mathbb{R} \mid f(\mathbf{x}) = \theta \forall \mathbf{x} \in \mathcal{X}\}$, where $\mathcal{X}$ is the feature space. Show that

$$\hat{f}(\mathbf{x}) = m^{-1} \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)$$

is the optimal constant model for the loss function above, where m^{-1} is the inverse function of m .

Solution to Question (a)

1. f is a **constant model**: $f(\mathbf{x}) = \theta$ for all $\mathbf{x}$.
2. The empirical risk can be formulated as:

$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^n \left(m(y^{(i)}) - m(f(\mathbf{x}^{(i)})) \right)^2 = \sum_{i=1}^n \left(m(y^{(i)}) - m(\theta) \right)^2.$$

3. $\mathcal{R}_{\text{emp}}(f)$ is **strictly convex** (because MSE loss and m is strictly monotone). So the minimum is unique, and can be computed by solving $\partial \mathcal{R}_{\text{emp}}(f) / \partial \theta = \mathbf{0}$.

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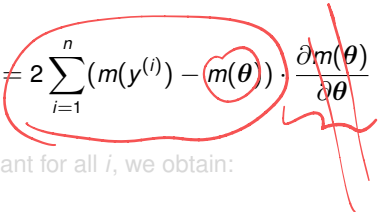
$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^n \left(m(y^{(i)}) - m(f(\mathbf{x}^{(i)})) \right)^2 = \sum_{i=1}^n \left(m(y^{(i)}) - m(\theta) \right)^2.$$

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Solution to Question (a): Continued

Goal: Compute the optimal θ by solving $\partial \mathcal{R}_{\text{emp}}(f)/\partial \theta = \mathbf{0}$.

1. Compute the derivative:

$$\frac{\partial \mathcal{R}_{\text{emp}}(f)}{\partial \theta} = 2 \sum_{i=1}^n (m(y^{(i)}) - m(\theta)) \cdot \frac{\partial m(\theta)}{\partial \theta} = 0$$


2. Using the fact that $\frac{\partial m(\theta)}{\partial \theta}$ is constant for all i , we obtain:

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2. Using the fact that $\frac{\partial m(\theta)}{\partial \theta}$ is constant for all i , we obtain:

$$\sum_{i=1}^n (m(y^{(i)}) - m(\theta)) = 0$$
$$\Rightarrow \sum_{i=1}^n m(y^{(i)}) = \sum_{i=1}^n m(\theta) \rightarrow n m(\theta)$$

Solution to Question (a): Continued

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1. Compute the derivative:

$$\frac{\partial \mathcal{R}_{\text{emp}}(f)}{\partial \theta} = 2 \sum_{i=1}^n (m(y^{(i)}) - m(\theta)) \cdot \frac{\partial m(\theta)}{\partial \theta} = 0$$

2. Using the fact that $\frac{\partial m(\theta)}{\partial \theta}$ is constant for all i , we obtain:

$$\begin{aligned} \sum_{i=1}^n (m(y^{(i)}) - m(\theta)) &= 0 \\ \Rightarrow \sum_{i=1}^n m(y^{(i)}) &= \sum_{i=1}^n m(\theta) \\ \Rightarrow m(\theta) &= \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \end{aligned}$$

Solution to Question (a): Continued

Goal: Compute the optimal θ by solving $\partial \mathcal{R}_{\text{emp}}(f)/\partial \theta = \mathbf{0}$.

1. Compute the derivative:

$$\frac{\partial \mathcal{R}_{\text{emp}}(f)}{\partial \theta} = 2 \sum_{i=1}^n (m(y^{(i)}) - m(\theta)) \cdot \frac{\partial m(\theta)}{\partial \theta} = 0$$

2. Using the fact that $\frac{\partial m(\theta)}{\partial \theta}$ is constant for all i , we obtain:

$$\sum_{i=1}^n (m(y^{(i)}) - m(\theta)) = 0$$

$$\Rightarrow \sum_{i=1}^n m(y^{(i)}) = \sum_{i=1}^n m(\theta)$$

$$\Rightarrow m(\theta) = \frac{1}{n} \sum_{i=1}^n m(y^{(i)})$$

$$\Rightarrow \theta^* = m^{-1} \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \triangleleft$$

Question (b)

$$\hat{f} = \theta^* = m^{-1} \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)$$

(b) Verify that the risk of the optimal constant model is $\mathcal{R}_L(\hat{f}) = (1 + \frac{1}{n})\text{Var}(m(y))$.

Recall that the risk of $\hat{f}$ is defined as

$$\mathcal{R}_L(\hat{f}) = \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x}))]$$

Solution to Question (b)

$$\begin{aligned}\mathcal{R}_L(\hat{f}) &= \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x}))] \\ &= \mathbb{E}_{xy}[(m(y) - m(\hat{f}(\mathbf{x})))^2]\end{aligned}$$

Solution to Question (b)

$$\mathbb{E}_{xy}[Q(y)]$$
$$= \int_y \int_x Q(y) p(x, y) dx dy$$

$$= \int_y Q(y) \underbrace{\int_x p(x, y) dx}_{p(y)} dy$$

$$= \int_y Q(y) p(y) dy$$

$$= \mathbb{E}_y[Q(y)]$$

$$m(\hat{f}(x)) = m\left(\frac{1}{n} \sum_{i=1}^n y^{(i)}\right)$$

$$\mathcal{R}_L(\hat{f}) = \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x}))]$$

$$= \mathbb{E}_{xy}[(m(y) - m(\hat{f}(\mathbf{x})))^2]$$

$$= \mathbb{E}_{xy} \left[\left(m(y) - \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right]$$

$$= \mathbb{E}_y[Q(y)] = \mathbb{E}_y \left[\left(m(y) - \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right]$$

Solution to Question (b)

$$\begin{aligned}\mathcal{R}_L(\hat{f}) &= \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x}))] \\&= \mathbb{E}_{xy}[(m(y) - m(\hat{f}(\mathbf{x})))^2] \\&= \mathbb{E}_{xy} \left[\left(m(y) - \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right] \\&= \mathbb{E}_{xy}[m(y)^2] - 2 \cdot \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] + \mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \right]\end{aligned}$$

Solution to Question (b): Continued

Now take a look at the second term: $-2 \cdot \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right]$.

$$\begin{aligned} & \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] \\ &= \frac{1}{n} \cdot \mathbb{E}_{xy} \left[m(y) \sum_{i=1}^n m(y^{(i)}) \right] \\ &= \frac{1}{n} \cdot \mathbb{E}_{xy}[m(y)] \mathbb{E}_{xy} \left[\sum_{i=1}^n m(y^{(i)}) \right] \end{aligned}$$

$$\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})]$$

$$\underbrace{m(y)}$$

$$\underbrace{\sum_{i=1}^n m(y^{(i)})}$$

Solution to Question (b): Continued

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Handwritten notes illustrating the independence of y and $y^{(i)}$:

$\textcircled{y} \perp\!\!\!\perp y^{(i)}$

$m(y) \perp\!\!\!\perp m(y^{(i)})$

$m(y) \perp\!\!\!\perp \sum_{i=1}^n m(y^{(i)})$

Solution to Question (b): Continued

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Why $\mathbb{E}_{xy} \left[\sum_{i=1}^n m(y^{(i)}) \right] = n \cdot \mathbb{E}_{xy}[m(y)]$?

- Linearity: pull $\mathbb{E}_{xy}$ inside the sum.
- (i) just labels different draws; all $y^{(i)}$ share y 's distribution, so $\mathbb{E}_{xy}[m(y^{(i)})] = \mathbb{E}_{xy}[m(y)]$.
- Sum of n equal terms $= n \cdot \mathbb{E}_{xy}[m(y)]$.

Analogy: expected sum of n dice rolls is $n \times 3.5$.

Solution to Question (b): Continued

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Solution (b): third term $\mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right]$

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$$\mathbb{E}_{xy} \left[(\dots + \dots + \dots) \right]$$
$$\underbrace{m(y^{(i)})^2} \quad \underbrace{2m(y^{(i)})m(y^{(j)})}$$

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$$\begin{aligned} & \mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right] \\ &= \frac{1}{n^2} \mathbb{E}_{xy} \left[\sum_{i,j=1}^n m(y^{(i)}) m(y^{(j)}) \right] \end{aligned}$$

Example ($n = 3$). Let $m_i := m(y^{(i)})$.

$$\left(\frac{m_1 + m_2 + m_3}{3} \right)^2 = \frac{1}{9} \sum_{i,j=1}^3 m_i m_j$$

Expand all 9 pairs (i, j) :

m_1^2	$m_1 m_2$	$m_1 m_3$
$m_2 m_1$	m_2^2	$m_2 m_3$
$m_3 m_1$	$m_3 m_2$	m_3^2

Group by color:

- ▶ **Diagonal** ($i = j$, n terms) $\rightsquigarrow \sum_i m_i^2$
- ▶ **Off-diag.** ($i \neq j$, $n(n-1)$ terms)
 $\rightsquigarrow \sum_i \sum_{j \neq i} m_i m_j$

Solution (b): third term $\mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right]$

$$\begin{aligned}
 & \mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right] \\
 &= \frac{1}{n^2} \mathbb{E}_{xy} \left[\sum_{i,j=1}^n m(y^{(i)}) m(y^{(j)}) \right] \\
 &= \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] \right. \\
 & \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [m(y^{(i)}) m(y^{(j)})] \right)
 \end{aligned}$$

▷ Split into **diagonal** ($i = j$, squared) and **off-diagonal** ($i \neq j$, cross) terms.

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 m_1^2 & m_1 m_2 & m_1 m_3 \\
 m_2 m_1 & m_2^2 & m_2 m_3 \\
 m_3 m_1 & m_3 m_2 & m_3^2
 \end{array}$$

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- ▶ **Off-diag.** ($i \neq j$, $n(n-1)$ terms) $\rightsquigarrow \sum_i \sum_{j \neq i} m_i m_j$

Solution (b): third term, cont.

Continuing from the grouping form:

$$\frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [\underline{m(y^{(i)})} \underline{m(y^{(j)})}] \right)$$

Solution (b): third term, cont.

Continuing from the grouping form:

$$\begin{aligned} & \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] \right. \\ & \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [m(y^{(i)}) m(y^{(j)})] \right) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] \right. \\ & \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [m(y^{(i)})] \mathbb{E}_{xy} [m(y^{(j)})] \right) \end{aligned}$$

Cross term: independence trick.

For $i \neq j$, $m(y^{(i)})$ and $m(y^{(j)})$ are *independent* RVs, so

$$\mathbb{E}_{xy} [m(y^{(i)}) m(y^{(j)})] = \mathbb{E}_{xy} [m(y^{(i)})] \mathbb{E}_{xy} [m(y^{(j)})].$$

Solution (b): third term, cont.

Continuing from the grouping form:

$$\begin{aligned} & \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] \right. \\ & \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [m(y^{(i)}) m(y^{(j)})] \right) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] \right. \\ & \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [m(y^{(i)})] \mathbb{E}_{xy} [m(y^{(j)})] \right) \\ &= \frac{1}{n^2} \left(n \mathbb{E}_{xy} [m(y)^2] + \underline{n(n-1)} \mathbb{E}_{xy} [m(y)]^2 \right) \end{aligned}$$

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Collapse by i.i.d.

- ▶ n **diagonal** terms, each = $\mathbb{E}_{xy} [m(y)^2]$
 $\Rightarrow n \mathbb{E}_{xy} [m(y)^2]$.
- ▶ $n(n-1)$ **off-diag.** terms, each = $\mathbb{E}_{xy} [m(y)]^2$
 $\Rightarrow n(n-1) \mathbb{E}_{xy} [m(y)]^2$.

Solution (b): third term, cont.

Continuing from the grouping form:

$$\begin{aligned}& \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy}[m(y^{(i)})^2] \right. \\& \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy}[m(y^{(i)})m(y^{(j)})] \right) \\&= \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy}[m(y^{(i)})^2] \right. \\& \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy}[m(y^{(i)})] \mathbb{E}_{xy}[m(y^{(j)})] \right) \\&= \frac{1}{n^2} \left(n\mathbb{E}_{xy}[m(y)^2] + n(n-1)\mathbb{E}_{xy}[m(y)]^2 \right) \\&= \underbrace{\frac{1}{n}\mathbb{E}_{xy}[m(y)^2]} + \underbrace{\left(1 - \frac{1}{n}\right)\mathbb{E}_{xy}[m(y)]^2}\end{aligned}$$

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For $i \neq j$, $m(y^{(i)})$ and $m(y^{(j)})$ are *independent* RVs, so

$$\mathbb{E}_{xy}[m(y^{(i)})m(y^{(j)})] = \mathbb{E}_{xy}[m(y^{(i)})] \mathbb{E}_{xy}[m(y^{(j)})].$$

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- ▶ $n(n-1)$ **off-diag.** terms, each = $\mathbb{E}_{xy}[m(y)]^2$
 $\Rightarrow n(n-1)\mathbb{E}_{xy}[m(y)]^2$.

Solution to Question (b): Continued

Combining the results so far, we get

$$\mathcal{R}_L(\hat{f}) = \mathbb{E}_{xy}[m(y)^2] - 2 \cdot \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] + \mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \right]$$

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Combining the results so far, we get

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$$\begin{aligned}& \mathbb{E}_y \left[\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] \cdot \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_y[m(y^{(i)})] \\ &= \frac{1}{n} \cdot n \cdot \mathbb{E}_y[m(y)]^2 = \mathbb{E}_y[m(y)]^2\end{aligned}$$

Solution to Question (b): Continued

Combining the results so far, we get

$$\begin{aligned}\mathcal{R}_L(\hat{f}) &= \mathbb{E}_{xy}[m(y)^2] - 2 \cdot \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] + \mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \right] \\&= \mathbb{E}_{xy}[m(y)^2] - 2\mathbb{E}_{xy}[m(y)]^2 + \frac{1}{n}\mathbb{E}_{xy}[m(y)^2] + \left(1 - \frac{1}{n}\right)\mathbb{E}_{xy}[m(y)]^2 \\&= \left(1 + \frac{1}{n}\right) \left(\mathbb{E}_{xy}[m(y)^2] - \mathbb{E}_{xy}[m(y)]^2 \right)\end{aligned}$$

(Handwritten red annotations: A circle around $\left(1 + \frac{1}{n}\right)$ and a bracket under the term $\mathbb{E}_{xy}[m(y)^2] - \mathbb{E}_{xy}[m(y)]^2$.)

Solution to Question (b): Continued

Combining the results so far, we get

$$\begin{aligned}\mathcal{R}_L(\hat{f}) &= \mathbb{E}_{xy}[m(y)^2] - 2 \cdot \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] + \mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \right] \\&= \mathbb{E}_{xy}[m(y)^2] - 2\mathbb{E}_{xy}[m(y)]^2 + \frac{1}{n}\mathbb{E}_{xy}[m(y)^2] + \left(1 - \frac{1}{n}\right)\mathbb{E}_{xy}[m(y)]^2 \\&= \left(1 + \frac{1}{n}\right) (\mathbb{E}_{xy}[m(y)^2] - \mathbb{E}_{xy}[m(y)]^2) \\&= \left(1 + \frac{1}{n}\right) \text{Var}(m(y)).\end{aligned}$$

$\text{Var}(X) = E[X^2] - (E[X])^2$

Why does $\mathbb{E}_x$ vanish?

- ▶ Constant model $\Rightarrow$ integrands depend on y only.
- ▶ Marginalize: $\int p(x, y) dx = p(y)$.
- ▶ Hence $\mathbb{E}_{xy}[m(y)] = \mathbb{E}_y[m(y)]$; $\text{Var}(m(y))$ is over y .

Question (c)

Derive that the risk minimizer f^* is given by $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y)|\mathbf{x}])$.

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The **unrestricted** space of all functions: $\mathcal{H} = \{f : \mathcal{X} \rightarrow \mathbb{R}\}$.

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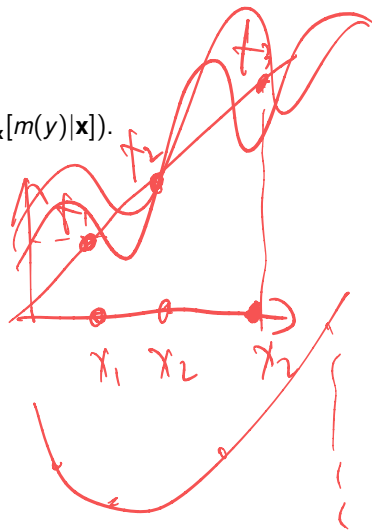
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Q2: What does “unrestricted” imply? Why?

What freedom does this give, and why does it matter?



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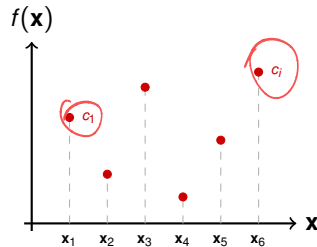
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What freedom does this give, and why does it matter?

A2

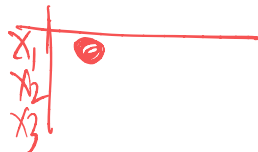
For **each** $\mathbf{x}$, $f(\mathbf{x})$ can take any $c \in \mathbb{R}$ *independently*. $\rightsquigarrow$

Point-wise: minimize risk separately at every $\mathbf{x}$.



Each $\mathbf{x}_i$ gets its own optimal c_i

— no shared functional form.



Solution to Question (c)

Recall (c): point-wise optimization $\Rightarrow$ decompose

$$p(x, y) = p(y|x)p(x)$$

$$\mathbb{E}_{xy}[\cdot] = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[\cdot | \mathbf{x}]].$$

Apply the law of total expectation to the risk:

$$\mathcal{R}_L(f) = \mathbb{E}_{xy}[L(y, f(\mathbf{x}))]$$

$$= \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f(\mathbf{x})))^2 | \mathbf{x}]].$$

To solve $f^*(\mathbf{x}) = \arg \min_c \mathbb{E}_{y|\mathbf{x}}[(m(y) - m(c))^2 | \mathbf{x}]$, same as (a):

$$f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) | \mathbf{x}]).$$

$$\int_{\mathbf{x}} \int_y Q(x, y) p(x, y) dy d\mathbf{x}$$

$$\int_{\mathbf{x}} \int_y Q(x, y) p(y|x) p(x) dy d\mathbf{x}$$

$$\mathbb{E}_{\mathbf{x}} \left[\int_y Q(x, y) p(y|x) dy \right]$$

$$\mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{y|x} [Q(x, y)] \right]$$

Solution to Question (c)

Recall (c): point-wise optimization $\Rightarrow$ decompose

$$\mathbb{E}_{xy}[\cdot] = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[\cdot | \mathbf{x}]].$$

$f(x) = c$ given x

Apply the law of total expectation to the risk:

$x \rightarrow c$

$$\begin{aligned}\mathcal{R}_L(f) &= \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] \\ &= \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[\underbrace{(m(y) - m(f(\mathbf{x})))^2}_{\downarrow} | \mathbf{x}]].\end{aligned}$$

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$$\frac{\partial}{\partial c} \mathbb{E}_{y|\mathbf{x}}[(m(y) - m(c))^2 | \mathbf{x}]$$

$$= 2 \mathbb{E}_{y|\mathbf{x}}[m(y) - m(c)]$$

$$= 2 \mathbb{E}_{y|\mathbf{x}}[m(y)] - 2 \mathbb{E}_{y|\mathbf{x}}[m(c)]$$

$$= 2 \mathbb{E}_{y|\mathbf{x}}[m(y)] - 2 \underline{m(c)}$$

$$m(c) = \mathbb{E}_{y|\mathbf{x}}[m(y)]$$

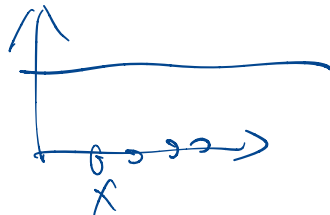
$$c = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y)])$$

Question (d)

(d): What is the optimal **constant** model in terms of the (theoretical) risk for the loss above and what is the risk?

Question (d)

$$f(x) = c$$

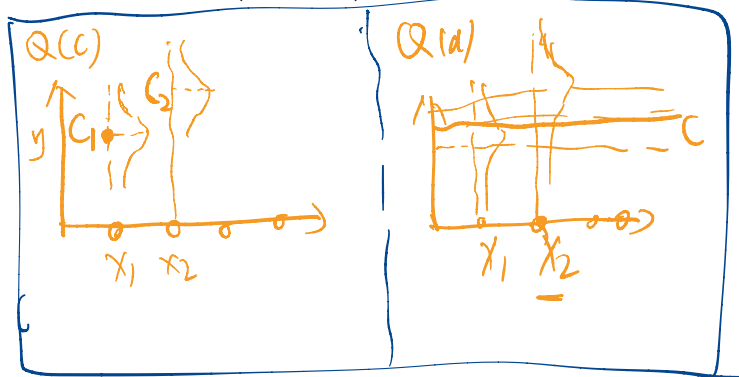


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Q1

Which hypothesis space $\mathcal{H}$?

$$E_{xy}$$



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Constant models: $\mathcal{H} = \{f : f(\mathbf{x}) = c \forall \mathbf{x}\}$.

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Empirical or theoretical risk?

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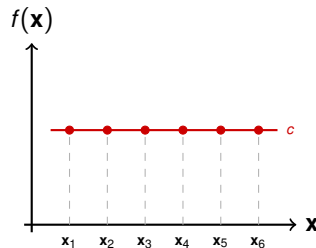
Constant models: $\mathcal{H} = \{f : f(\mathbf{x}) = c \forall \mathbf{x}\}$.

Q2

Empirical or theoretical risk?

A2

The **theoretical** risk $\mathcal{R}_L(f) = \mathbb{E}_{xy}[L(y, f(\mathbf{x}))]$ (not empirical as in (a)).



One c for all $\mathbf{x}$.

Solution to Question (d)

The (theoretical) risk for a constant model $\bar{f}(\mathbf{x}) = c$ is:

$$\begin{aligned}\mathcal{R}_L(\bar{f}) &= \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - m(\bar{f}(\mathbf{x})))^2]] \\&= \int_y \int_{\mathbf{x}} (m(y) - m(\bar{f}(\mathbf{x})))^2 p(\mathbf{x}, y) d\mathbf{x} dy \\&= \int_y \int_{\mathbf{x}} (m(y) - m(c))^2 p(\mathbf{x}, y) d\mathbf{x} dy \\&= \int_y (m(y) - m(c))^2 p(y) dy \\&= \mathbb{E}_y [(m(y) - m(c))^2]\end{aligned}$$

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$(m(y) - m(c))^2$ is $\mathbf{x}$ -free $\Rightarrow$ pull out of $\int_{\mathbf{x}}$,
then marginalize:

$$\begin{aligned}&\int_{\mathbf{x}} (m(y) - m(c))^2 p(\mathbf{x}, y) d\mathbf{x} \\&= (m(y) - m(c))^2 \underbrace{\int_{\mathbf{x}} p(\mathbf{x}, y) d\mathbf{x}}_{\substack{\text{red circle} \\ = p(y)}}\end{aligned}$$

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$$\begin{aligned}m(c^*) &= \mathbb{E}_y [m(y)] \\c^* &= m^{-1}(\mathbb{E}_y [m(y)])\end{aligned}$$

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Therefore, the optimal constant model is

$$\bar{f}(\mathbf{x}) = c = m^{-1}(\mathbb{E}_y[m(y)])$$

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Solution to Question (d): Continued

The **risk** of the optimal constant model $\bar{f}(\mathbf{x}) = c = m^{-1}(\mathbb{E}_y[m(y)])$:

$$\begin{aligned}\mathcal{R}_L(\bar{f}) &= \mathbb{E}_{xy}[(m(y) - m(\bar{f}(\mathbf{x})))^2] \\ &= \mathbb{E}_y[(m(y) - \mathbb{E}_y[m(y)])^2] \\ &= \text{Var}(m(y)).\end{aligned}$$

$$\begin{aligned}m(\bar{f}(\mathbf{x})) \\ &= m(c) \\ &= m(m^{-1}(\mathbb{E}_y[m(y)])) \\ &= \mathbb{E}_y[m(y)]\end{aligned}$$

Solution to Question (d): Continued

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Solution to Question (d): Continued

$$p(y') dy' = P(y) dy$$

$$y' = m(y)$$

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$$\mathbb{E}_{y'}[(y' - \mathbb{E}_{y'}[y'])^2]$$

m

Definition of variance: for a random variable Z , $\text{Var}(Z) = \mathbb{E}[(Z - \mathbb{E}[Z])^2]$. Here $Z = m(y)$.

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Question (e)

(e): Recall the decomposition of the Bayes regret into the estimation and the approximation error. Show that the former is $\frac{1}{n} \text{Var}(m(y))$, while the latter is $\text{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])$ for the optimal constant model $\hat{f}(\mathbf{x})$ if $\mathcal{H}$ consists of the constant models.

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Q1: What is the Bayes regret, and what are its two errors?

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Q1: What is the Bayes regret, and what are its two errors?

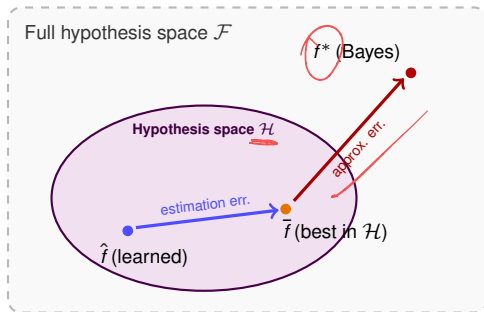
A1: Bayes regret = $\mathcal{R}_L(\hat{f}) - \mathcal{R}_L^*$, splits as:

Estimation error:

$$\mathcal{R}_L(\hat{f}) - \inf_{f \in \mathcal{H}} \mathcal{R}_L(f)$$

Approximation error:

$$\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*$$



Solution to Question (e): Roadmap

Three risk minimizers from earlier parts, connected by the two errors:

From (a)

$$\hat{f}(\mathbf{x}) = m^{-1} \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)$$

$$\mathcal{R}_L(\hat{f}) = \left(1 + \frac{1}{n}\right) \text{Var}(m(y))$$

$\therefore n$

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estimation
err.

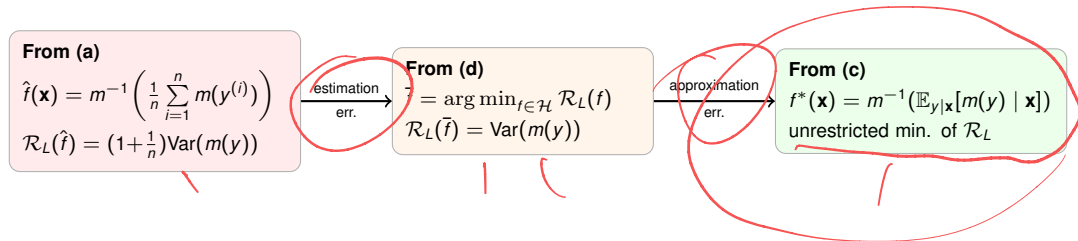
From (d)

$$\bar{f} = \arg \min_{f \in \mathcal{H}} \mathcal{R}_L(f)$$

$$\mathcal{R}_L(\bar{f}) = \text{Var}(m(y))$$

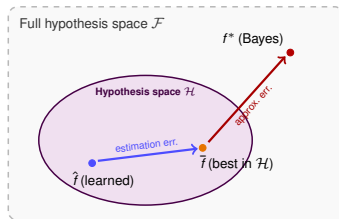
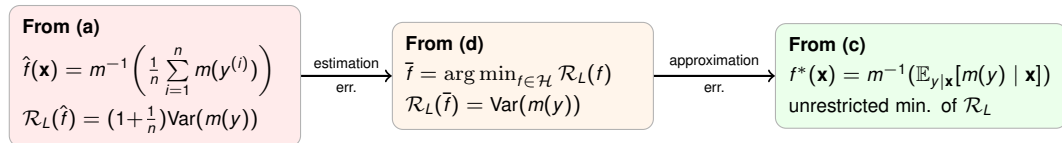
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Solution to Question (e): Roadmap

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Solution (e): Part 1 of 2 — Estimation Error

$$\begin{aligned}\mathcal{R}_L(\hat{f}) - \inf_{f \in \mathcal{H}} \mathcal{R}_L(f) \\&= \mathcal{R}_L(\hat{f}) - \mathcal{R}_L(\bar{f}) \\&= \left(1 + \frac{1}{n}\right) \text{Var}(m(y)) - \text{Var}(m(y)) \\&= \boxed{\frac{1}{n} \text{Var}(m(y))}\end{aligned}$$

Solution (e): Part 1 of 2 — Estimation Error

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$\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) = \mathcal{R}_L(\bar{f})$, since $\bar{f}$ is the in-class minimizer (from (d)).

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$\mathcal{R}_L(\hat{f}) = (1 + \frac{1}{n}) \text{Var}(m(y))$ from (a);
 $\mathcal{R}_L(\bar{f}) = \text{Var}(m(y))$ from (d).

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Solution (e): Part 2 of 2 — Approximation Error

$$\begin{aligned} & \inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^* \\ &= \mathcal{R}_L(\bar{f}) - \mathcal{R}_L(f^*) \\ &= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - m(f^*(\mathbf{x})))^2 \mid \mathbf{x}]] \\ &= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])^2 \mid \mathbf{x}]] \\ &= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\text{Var}(m(y) \mid \mathbf{x})] \\ &= \boxed{\text{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])} \end{aligned}$$

Solution (e): Part 2 of 2 — Approximation Error

$$\begin{aligned} & \inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^* \\ &= \mathcal{R}_L(\bar{f}) - \mathcal{R}_L(f^*) \\ &= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - m(f^*(\mathbf{x})))^2 \mid \mathbf{x}]] \\ &= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])^2 \mid \mathbf{x}]] \\ &= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\text{Var}(m(y) \mid \mathbf{x})] \\ &= \boxed{\text{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])} \end{aligned}$$

Solution (e): Part 2 of 2 — Approximation Error

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From (d): $\mathcal{R}_L(\bar{f}) = \text{Var}(m(y))$.

Def.: $\mathcal{R}_L(f^*) = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[(m(y) - m(f^*(\mathbf{x})))^2 \mid \mathbf{x}]]$.

Solution (e): Part 2 of 2 — Approximation Error

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$$\mathbb{E}_{\mathbf{x}} [\text{Var}(m(y) | \mathbf{x})]$$

From (d): $\mathcal{R}_L(\bar{f}) = \text{Var}(m(y))$.

Def.: $\mathcal{R}_L(f^*) = \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - m(f^*(\mathbf{x})))^2 | \mathbf{x}]]$.

From (c): $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) | \mathbf{x}])$,
so $m(f^*(\mathbf{x})) = \mathbb{E}_{y|\mathbf{x}}[m(y) | \mathbf{x}]$.

Solution (e): Part 2 of 2 — Approximation Error

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Solution (e): Part 2 of 2 — Approximation Error

$$\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*$$

$$= \mathcal{R}_L(\bar{f}) - \mathcal{R}_L(f^*)$$

$$= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - m(f^*(\mathbf{x})))^2 | \mathbf{x}]]$$

$$= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - \mathbb{E}_{y|\mathbf{x}}[m(y) | \mathbf{x}])^2 | \mathbf{x}]]$$

$$= \text{Var}(m(y)) - \mathbb{E}_{\mathbf{x}} [\text{Var}(m(y) | \mathbf{x})]$$

$$= \boxed{\text{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) | \mathbf{x}])}$$

Var of Exp.

Exp. of Var

From (d): $\mathcal{R}_L(\bar{f}) = \text{Var}(m(y))$.

Def.: $\mathcal{R}_L(f^*) = \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - m(f^*))^2 | \mathbf{x}]]$.

From (c): $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) | \mathbf{x}])$,
so $m(f^*(\mathbf{x})) = \mathbb{E}_{y|\mathbf{x}}[m(y) | \mathbf{x}]$.

Law of total variance: $\text{Var}(Y) = \mathbb{E}_X [\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X])$.